PRM393 — FLUTTER MOBILE DEVELOPMENT

LAB 11.5 — TEST SUITE SUMMARY REPORT

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1. Executive Summary

The Taskly Flutter application has been thoroughly validated through a complete multi-layered automated test suite. The suite spans Unit Tests (pure Dart domain logic), Widget Tests (component rendering & local user interactions), Navigation Tests (screen routing & parameter passing), and Integration Tests (full end-to-end user workflows). All 14 tests execute cleanly with a 100% pass rate.

2. Test Suite Execution Summary Table

Test Category	Test File	Tests	Status
Unit Testing (Model)	test/unit/task_model_test.dart	3	PASSED (100%)
Unit Testing (Repo)	test/unit/task_repository_test.dart	5	PASSED (100%)
Widget Testing (UI)	test/widget/task_list_widget_test.dart	3	PASSED (100%)
Navigation Testing	test/widget/task_navigation_test.dart	2	PASSED (100%)
Integration Testing	test/integration/task_integration_test.dart	1	PASSED (100%)
TOTAL SUITE	**5 Test Files**	**14**	**ALL 14 PASSED**

3. Detailed Breakdown by Test Type

• **Unit Tests (8 Tests):** Adhered strictly to the AAA (Arrange-Act-Assert) pattern. Validated that Task defaults completed to false, that toggle() reversibly switches true/false, and copyWith() returns independent clones. Validated TaskRepository addTask(), deleteTask(), updateTask(), and edge-case handling for missing keys.

• **Widget Tests (3 Tests):** Simulated UI rendering using pumpWidget(), enterText(), tap(), and pump(). Validated the empty state banner ("No tasks yet. Add one!"), task creation via text input & submit button, and multiple task rendering.

• **Navigation Tests (2 Tests):** Seeded tasks, pumped TaskListScreen, simulated tapping an item, executed pumpAndSettle(), and verified arrival at TaskDetailScreen with title "Task Detail" and TextField key "detailTitleField". Also verified pop return navigation.

• **Integration Tests (1**

Test): Simulated end-to-end task workflow: Add "Original title" -> Open detail screen -> Edit text to "Updated title" -> Save -> Verify "Updated title" in list and "Original title" removed.

4. Behaviors Validated

' State encapsulation: TaskRepository mutations trigger clean, isolated UI refreshes via ChangeNotifier.
' User Input handling: Auto-clearing input field upon submission, whitespace trimming.
' Navigation data integrity: Task state flows bidirectionally between list and detail screens.
' Defensive logic: Safely handles empty lists, non-existent task IDs, and repeated toggle operations.

5. Known Limitations & Next Steps

1. In-Memory Storage: Tasks reset upon app kill. Recommended enhancement: Integrate Hive / SQLite persistence.
2. Remote Synchronization: No cloud sync backend. Recommended enhancement: Add Firebase Cloud Firestore.
3. Semantic Accessibility: Additional Semantics labels should be provided for screen reader tools.